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02-06 NOV 2026
MEXICAN INTERNATIONAL CONFERENCE ON ARTIFICIAL INTELLIGENCE
|ITESM Campus Chihuahua

Paper #12 · Oral

Forecasting Weekly Depression Incidence in Mexico:
A Multi-Model Framework with Auditable Per-Series Model Selection

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Independent research; affiliations do not imply institutional endorsement.

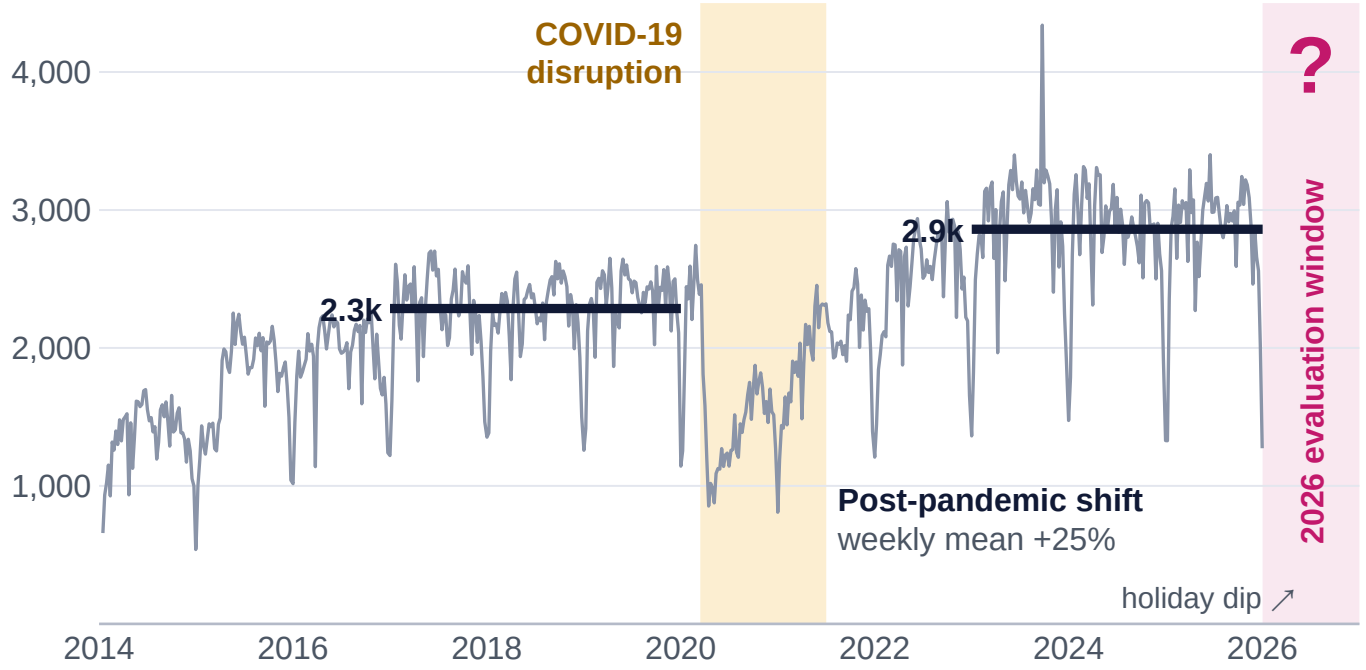
When historical scores meet recorded 2026 forecasts

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01 THE PROBLEM

Depression is counted weekly, but only in hindsight

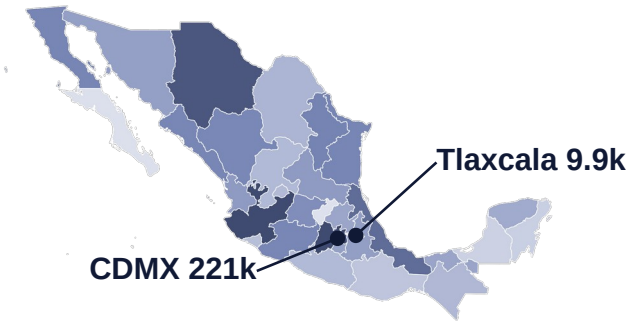
Weekly reported cases, national ICD-10 F32, SINAVE 2014–2025



~150k
cases per year
2023–2025

74%
are women
annual totals by sex

~22× **spread across states**



9.9k — 221k (log)

Source: SINAVE bulletins [1]; reported counts, not prevalence. Lines: weekly means 2017–19, 2023–25. Curve omits 2014-W02 and 2016-W20 (flagged partial).

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02 THE FRAMEWORK

Four candidates per series, one auditable rule

SINAVE bulletins

Weekly ICD-10 F32 counts, 2014–2025 (626 weeks)

111 targets: 32 states, 4 regions, national; general, women, men

99 reported directly

Four candidates

- **DeepAR** [3] probabilistic recurrent net
- **Prophet** [2] trend + seasonality
- **Ensemble** [4] Prophet + XGBoost blend
- **Stacking** [5, 6] Prophet, ETS, LightGBM

Deterministic selection rule

1. **Lowest historical sMAPE**
2. **Within 5% relative: lowest MASE**
3. **MASE tie: lowest RMSE**

Recorded forecast

52 weeks ahead

Audit log

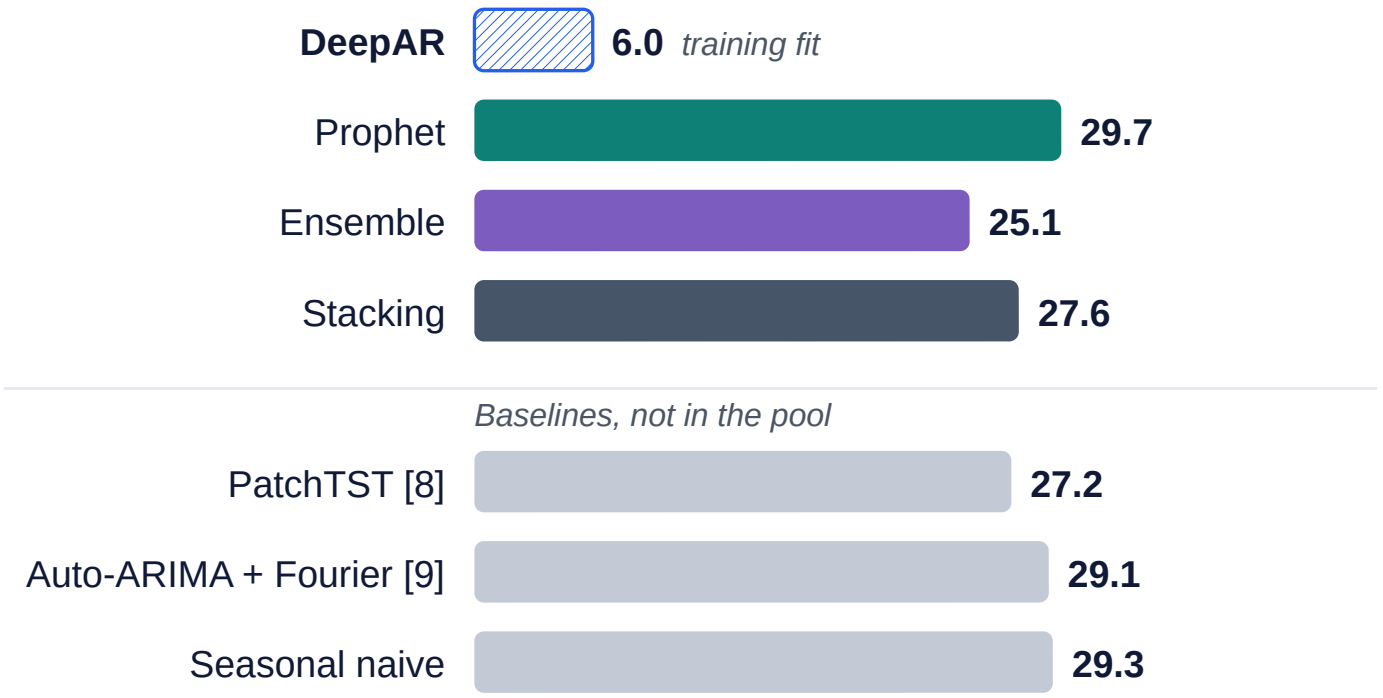
which model, on what evidence

Selection evidence: per-model historical scores (paper Table 1); protocols differ by model. MASE as in [7]. The sparse fallback never triggered (0 of 111).

03 SELECTION EVIDENCE

Historical selection used non-comparable scores

Median historical sMAPE (%) 111 series · lower is better



Series the rule assigns to DeepAR

107 / 111

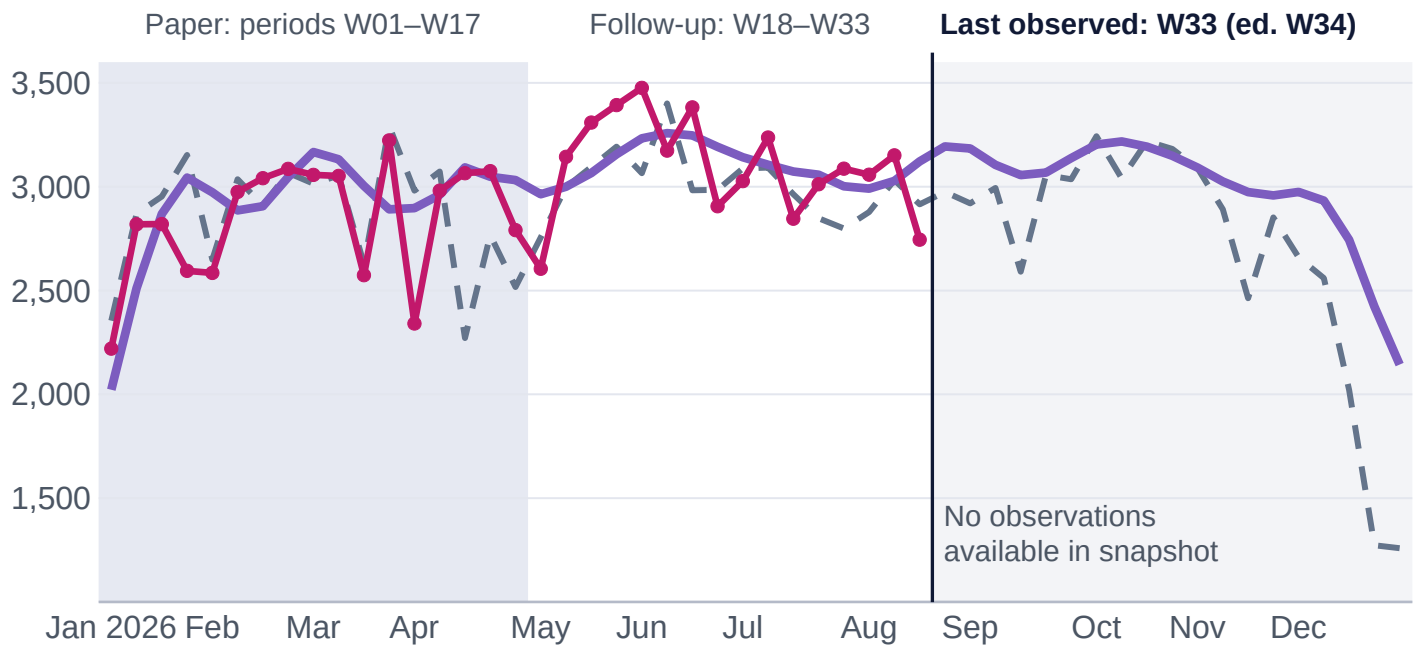
96.4%, including every state series

Caveat: Historical state DeepAR code uses training fit. Scores are not comparable.

Paper Table 1 (labelled in-sample selection metrics). DeepAR's 6.0 is the median over all 111 series. These figures select models; they do not measure accuracy.

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Recorded trajectories and available follow-up



Paper editions W02–W18	sMAPE (%)
Ensemble	7.40%
52-week naive	7.33%

Follow-up editions W19–W34	sMAPE (%)
Ensemble	5.91%
52-week naive	6.24%

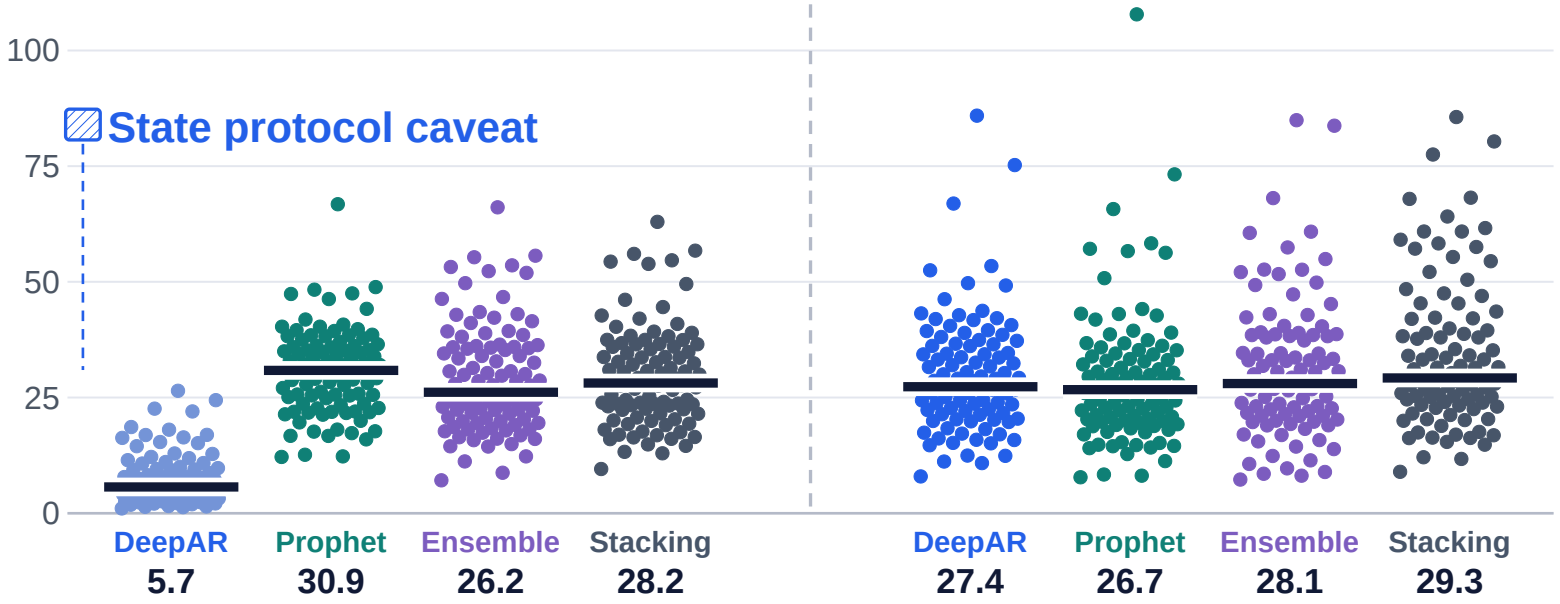
Forecast periods mapped by rule; execution origin unverified.

sMAPE. Cumulative deviation, Ensemble / naive: paper +2.45% / +0.6%; follow-up +0.17% / −2.95%. Snapshot: edition W34. Descriptive.

DeepAR’s historical margin disappears in 2026

Per-series sMAPE (%) 99 series, one dot each · bar = median · lower is better

Historical · protocols differ | **2026 recorded scores** · W03–W18



DeepAR’s historical margin

≈20 pts

2026 spread, all four

< 3 pts

No model is best everywhere



series won in 2026: Prophet, DeepAR, Ensemble, Stacking

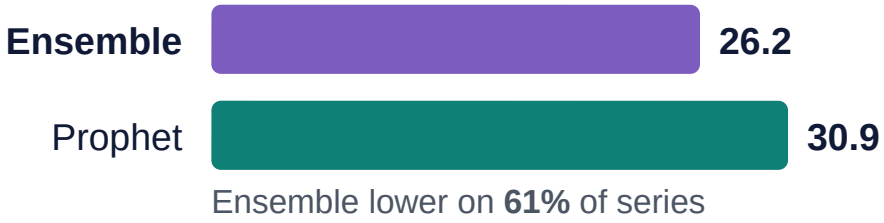
Historical protocols differ by model. Post hoc and exploratory: targets overlap; paired tests in backup B2.

06 WITHOUT DEEPAR

Excluding DeepAR, the two rankings differ

Median sMAPE (%) Ensemble vs. Prophet · same 99 series · lower is better

Historical held-out windows



Ensemble lower on **61%** of series

2026 recorded scores W03–W18



Prophet lower on **60%** of series

Same lowest-error model, historical vs. 2026

Prophet, Ensemble, Stacking · 99 series

34% **observed**
34 of 99 series

31% **reference**
from each model's win frequencies

Descriptive reference, not a formal chance test.

Point rankings; post hoc and exploratory. Averaging each state's three targets weakens the reversal (backup B2).

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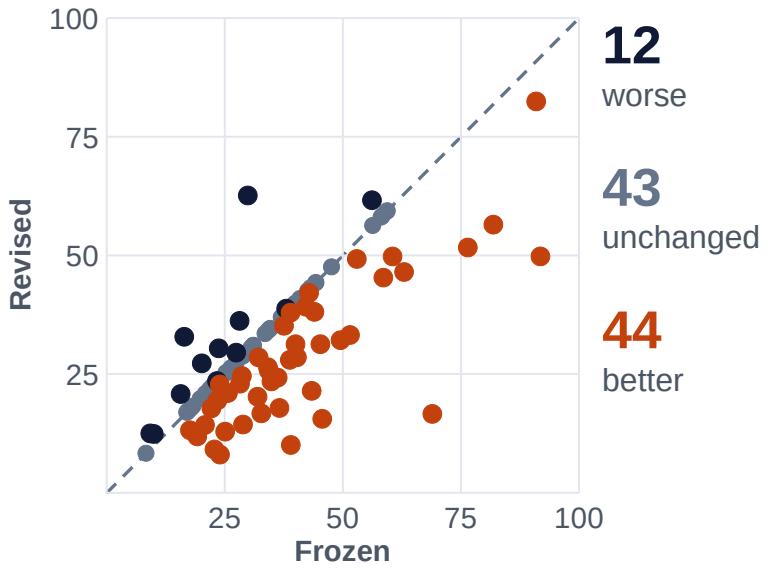
07 REVISION REPLAY

Per-series revision beat the frozen and global picks

W03–W11 re-select: same rule, prior weeks only

W12–W18 evaluation: excluded from reselection

Each dot is one series sMAPE, W12–W18



12 worse

43 unchanged

44 better

Policies median sMAPE, W12–W18

Frozen assignment	32.7
Global re-selection	32.7
Per-series revision	**28.5**

Comparators

Prophet only	27.6
Pool without DeepAR	26.4

−4.2 pts difference of policy medians

Caveat

Simple alternatives have lower medians.

Revision beat the frozen and global picks, not every alternative.

Revised vs. frozen: 44 better, 12 worse, 43 equal; lower medians at every split W06–W14 and in each stratum (exploratory).

TAKEAWAYS

Selection is a signal, not a guarantee

A reproducible audit of recorded forecasts and a time-ordered replay of selection.

1

Check that evidence is like-for-like

The released assignment selects DeepAR for all 96 state targets, using non-comparable historical evidence.

2

Test revision against simple alternatives

Replay W12–W18: revised 28.5; pool without DeepAR 26.4. Follow-up W19–W34: archived median 31.0; naive 27.0.

3

Know the limits

One condition. Revision: 7 weeks; follow-up: 16. Overlapping targets; weekly refitting and transfer untested.

THANK YOU

Questions?

Selection is a signal, not a guarantee.

Forecasting Weekly Depression Incidence in Mexico: A Multi-Model Framework with Auditable Per-Series Model Selection

To appear in Advances in Soft Computing (MICAI 2026), Springer LNCS 17149

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github.com/EpiForecastMX/per-series-model-selection2026

epiforecast.mx/micai2026
follow-up, data and slides

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SOURCES

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This work. Rebull-Saucedo, J.A., Pérez-Nava, J.C., Sánchez-Salazar, L.G., Barceló-Alonso, G., Pérez-Hernández, R.: Forecasting weekly depression incidence in Mexico: a multi-model framework with auditable per-series model selection. In: Advances in Soft Computing, MICAI 2026, LNCS 17149, Springer (to appear).

The selection rule, exactly

Per series, with recorded historical sMAPE, MASE, and RMSE per model:

1. If the best sMAPE beats the second by more than 5% (relative), pick it.
2. Otherwise, among models with sMAPE ≤ 1.05 × best, pick the lowest MASE.
3. If MASE is tied, pick the lowest RMSE.

\+ Fewer than 5 cases in the 52 weeks before the origin: use the regional model (past data only; never triggered here, 0 of 111).

Each choice and its metrics are stored with the forecast: the audit log.

Selection evidence

Per-model historical backtests (paper Table 1), not one shared protocol.

Prophet, Ensemble, Stacking: held-out windows, with different protocols.

Historical state DeepAR code skips CV, trains on the full series, then scores that fit (commit 60d6ea0f; changed in 9fdd6dca). The exact published runs are not independently linked.

The paper reports a data cutoff of 2025-W53; execution dates are not independently verified.

Statistical tests and uncertainty

Comparison	99 targets: raw / Holm	32 states: nominal
Historical, Ensemble vs. Prophet	0.020 / 0.060	0.15
2026, Prophet vs. Ensemble	0.007 / 0.035	0.06
2026, Prophet vs. DeepAR	0.070 / 0.14	0.24
Revision vs. frozen, split W11	< 0.001 / < 0.001	< 0.001

Exploratory, post hoc; targets overlap and Holm does not remove dependence. Not significant ≠ equivalent. Two-sided Wilcoxon, zsplit; Holm within six model pairs (nine for revision).

National, W03–W18, recomputed

Mean Poisson deviance: Ensemble 25.7, Prophet 32.3, DeepAR 46.0, Stacking 97.8.

Reported in the paper only

Diebold–Mariano tests; the 80% band covered 10 of 17 weeks. Inputs not released.

Limitations

One condition

Depression (F32) only; transfer is untested.

Training-fit evidence

Historical state DeepAR code uses training fit; run provenance is unverified.

Unequal search budget

Prophet had the broadest search; tuning effect untested.

Partial-year evaluation

Paper windows plus 16 follow-up periods; full-year evaluation remains pending.

Provenance

Forecast execution dates not independently verified.

Uncertainty

Paper-reported 80% band coverage: 58.8%; revision does not calibrate it.

Reported incidence

Models learn reporting and disease together.

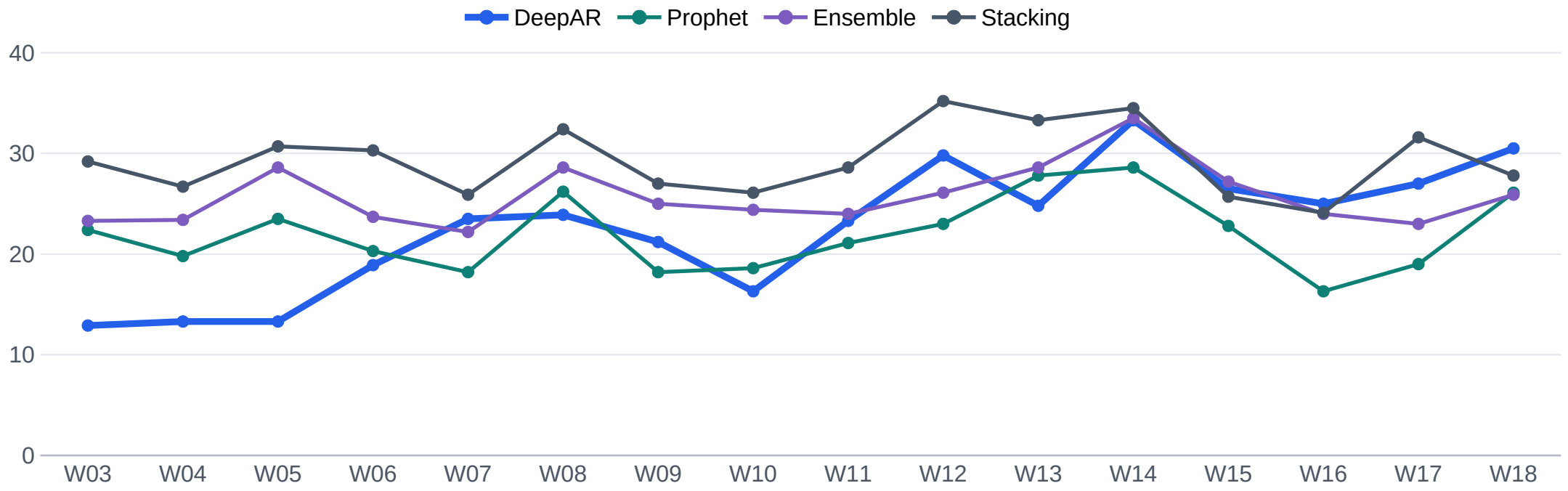
No external validation

Only SINAVE / Mexico; a research prototype.

B4 BACKUP · HORIZON

DeepAR leads the first four weeks; later ranks vary

Median per-week sMAPE (%) across the 99 series 2026 bulletin weeks · lower is better



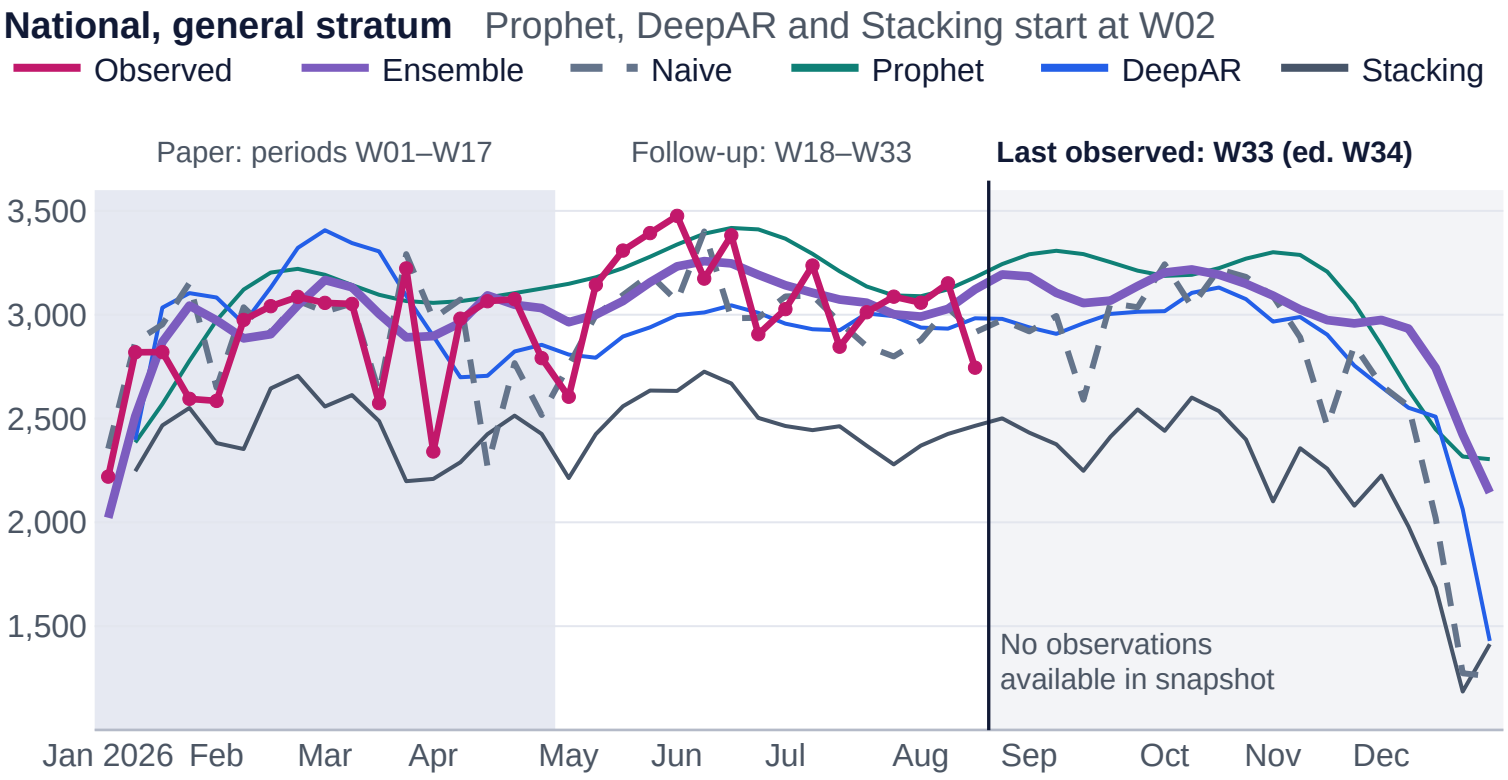
Revision chooses among frozen 52-week paths; horizon effects may contribute to its gain (not isolated). Recomputed from the public files.

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B5 BACKUP · FOLLOW-UP

Recorded 2026 paths: all four engines

National, general stratum Prophet, DeepAR and Stacking start at W02



99 targets, editions W19–W34
median sMAPE · targets below naive

Prophet	**23.4**	70/99
Ensemble	**25.2**	71/99
DeepAR	**31.0**	52/99
Stacking	**30.6**	51/99
Archived	**31.0**	55/99
Naive	**27.0**	

Forecast periods mapped by rule; execution origin unverified.

Descriptive; targets overlap. W01 reproduces the paper; legacy mapping is ambiguous. Archived = frozen assignment (96 DeepAR).